Experiment 01 — The 5-layer citation resolver

arxivkg · preprocessing · phase 3 (parse)

What we did

Phase-3 `parse` turns each in-text citation into a resolved arXiv id through a **cheapest-first, early-terminating** cascade: `link` → `text` (regex arXiv-id) → `doi` → `jref` → `title-exact` → `title-fuzzy` (with `title-exact-noyr` a year-relaxed variant; `rejected-*`/`none` for misses). We read `citation.method` from the single Postgres store and characterize each layer's contribution and how resolution varies by field and era.

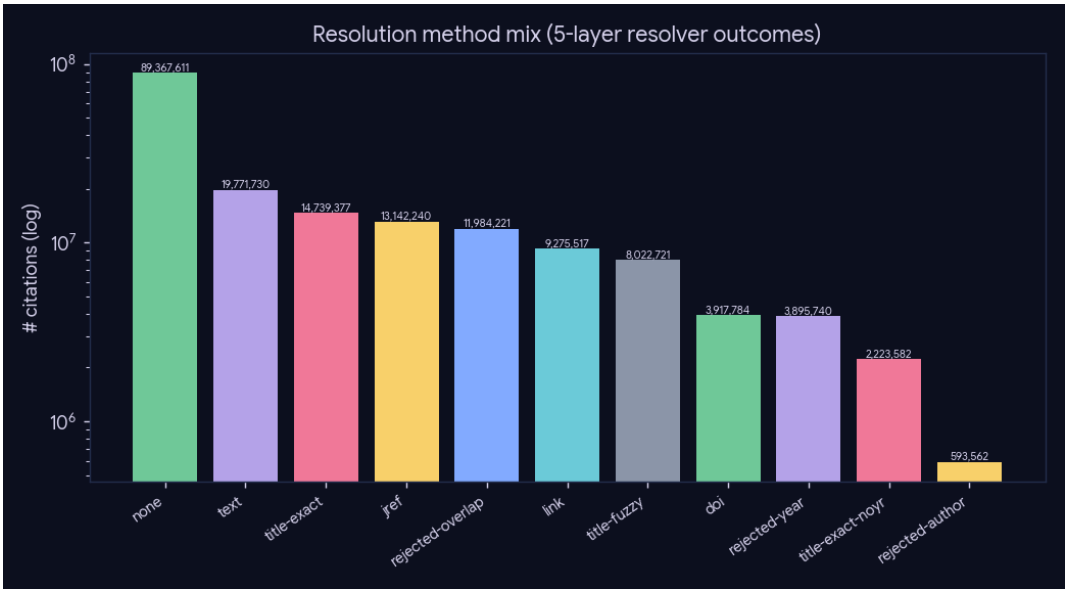
Method

Pure SQL aggregations over the 177M-row `citation` table; the heavy join is cached to `cache/summary.json`. No model — this measures the built artifact, not a runtime path.

Results

Overall resolution rate **40.18%** (71,092,951 / 176,934,085 occurrences). The cheap exact layers (`text`, `title-exact`, `jref`, `link`) carry ∼80 % of all resolutions; the `title-fuzzy` tail is bounded at ∼11 %.

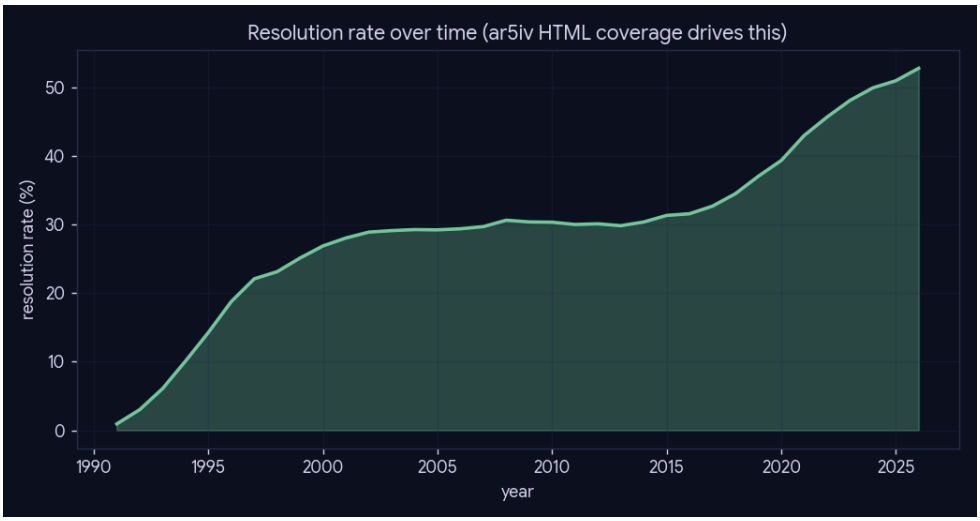
layer (method)	resolved	share of resolved
text	19,771,730	27.8%
title-exact	14,739,377	20.7%
jref	13,142,240	18.5%
link	9,275,517	13.0%
title-fuzzy	8,022,721	11.3%
doi	3,917,784	5.5%
title-exact-noyr	2,223,582	3.1%



By field and era

Resolution rate varies by field (HTML/LaTeX availability + citation style) and rises over time as ar5iv HTML coverage improves.





Conclusion

The cheapest-first resolver resolves **40 %** of all citation occurrences corpus-wide, the high-precision exact methods doing the bulk and the fuzzy layer adding a bounded ∼11 % tail — recall without leaning on the riskiest matcher. The remaining ∼60 % `none`/`rejected-*` is dominated by citations to non-arXiv venues with no arXiv id to resolve to. `rejected-*` rows are kept deliberately so the fuzzy layer's precision/recall trade-off is visible, not silently dropped.